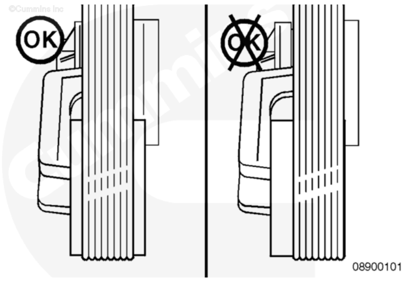
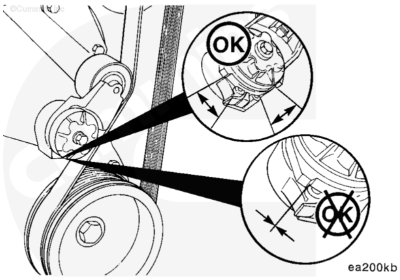
Trainings

Maintenance Check

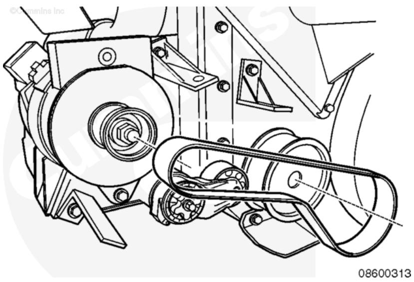
Check the location of the drive belt on the belt tensioner pulley. The belt should be centered on , or centered close to the middle of the pulley. Unaligned belts, either too far forward or backward, can cause belt wear, belt roll-off failures or increase uneven tensioner bushing wear.



With the belt on, check that neither tensioner arm stops are in contact with the spring casing stop. If either stop is touching, the drive belt **must** be replaced. After replacing the belt, if the tensioner are stops are still in contact with the spring casing stop, replace the tensioner.

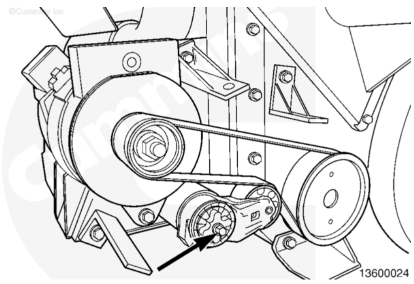


Remove the belt.



⚠ CAUTION ⚠

Do not increase the torque on engines that have grade 9.8 capscrews installed. Increasing torque can cause damage to the belt tensioner.



Engines that use grade 9.8 alternator tensioner mounting capscrews, must be tightened to the correct torque value.

If the alternator belt tensioner is removed and the mounting capscrew is grade 9.8, it must be replaced with a 10.9 capscrew.

Check the torque of the tensioner capscrew.

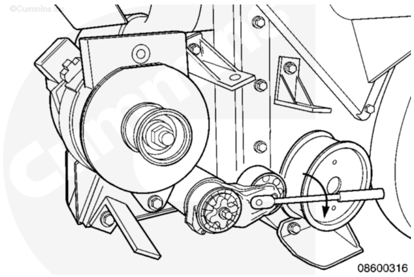
The torque for a grade 9.8 capscrew is:

Torque Value: 45 n•m [33 ft-lb]

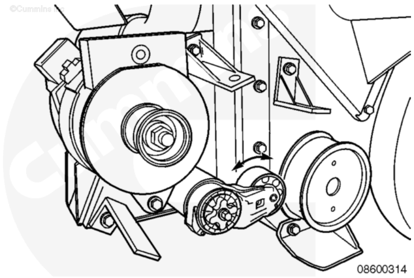
The torque for a grade 10.9 capscrew is:

Torque Value: 65 n•m [48 ft-lb]

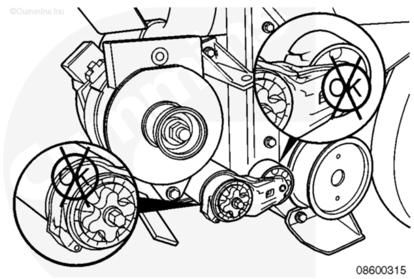
After checking the torque, use a breaker bar with a 1/2-inch ratchet to rotate the tensioner slowly away from the area of belt contact. If the arm rotates with any roughness or hesitancy, replace the tensioner.



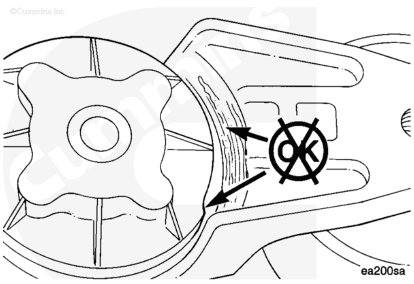
With the belt removed, check to be sure that the tensioner pulley rotates freely.



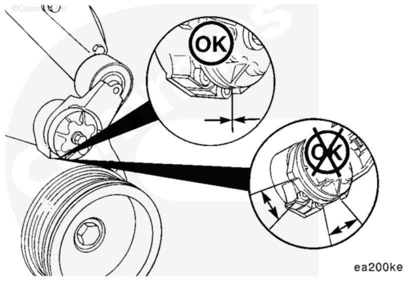
Check the tensioner arm, pulley, and stops for cracks. If any cracks are noticed, the tensioner **must** be replaced.



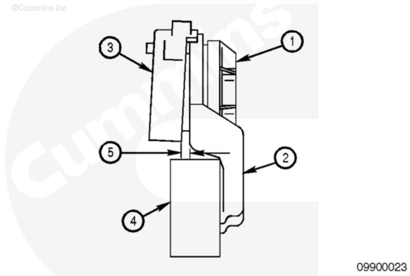
Inspect the tensioner for evidence of the tensioner arm contacting the tensioner cap. If there is evidence of the two areas making contact, the pivot tube bushing has failed and the tensioner **must** be replaced.



With the belt removed, check that the tensioner arm stop is in contact with the spring case stop. If these two are **not** touching, the tensioner **must** be replaced.



Measure the clearance between the tensioner spring case and the tensioner arm to verify tensioner wear-out and uneven bearing wear. If the clearance exceeds 3 mm [0.12 in] at any point, the tensioner has failed and **must** be replaced as a complete assembly. Experience has revealed that tensioners generally will show a larger clearance gap near the lower portion of the spring case, resulting in the upper portion rubbing against the tensioner arm. **Always** replace the belt when a tensioner is replaced. The belt tensioner assembly contains the following.



1. Tensioner cap
2. Tensioner arm
3. Spring case
4. Tensioner pulley
5. Clearance gap

Last Modified: 20-Jul-2018